Topic 28 — Environmental Research & Institutions

Covers syllabus (click to expand)

BSI | ZSI | FSI | FRI | WII | NBA | SBB | BMC | NBPGR

Sources baked in: MoEFCC; ICAR; ICFRE; Biological Diversity Act 2002; ISFR 2023; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — BSI/ZSI trap; ISFR→FSI; NBA Chennai; BMC→PBR; NBPGR Delhi **Last verified:** August 2026

Current Affairs (this topic)

- FRI / FSI / WII / IIRS function traps are evergreen Dehradun (देहरादून) CA colour.
- Do not swap ISFR (FSI) with FRI research mandate.

28.1 Botanical Survey of India (BSI) (भारतीय वनस्पति सर्वेक्षण)

Identity: 1890 | Kolkata | Plants only | Flora of India | Red Data Book (plants)

- **BSI** is the national organisation for survey, taxonomy, and documentation of **Indian plant diversity**.
- Established **1890**; headquarters **Kolkata** (Central National Herbarium (पादप संग्रहालय), Shibpur, Howrah); under **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
- Publishes ***Flora of India*** and state/regional floras; maintains herbarium network.
- Publishes the **Red Data Book of Indian Plants** — national plant threat list (distinct from IUCN Red List).
- Discovers and describes new plant species — especially Western Ghats (पश्चिमी घाट), Northeast, Andamans.
- **Does not** survey animals (ZSI), map forest cover (FSI), or store crop gene banks (NBPGR).

Logic: BSI = **plants only**. "BSI surveys animals" = **FALSE**.

28.2 Zoological Survey of India (ZSI) (भारतीय प्राणी सर्वेक्षण)

Identity: 1916 | Kolkata | Animals only | Fauna of India | Red Data Book (animals)

- **ZSI** documents **Indian animal (faunal) diversity** — taxonomy and inventories.
- Established **1916**; HQ **Kolkata**; under **MoEFCC**.
- Publishes ***Fauna of India*** series and the **Red Data Book of Indian Animals**.

- Maintains type specimens; scientific data informs WPA Schedule amendments.
- **ZSI ≠ WII:** ZSI = species inventory/taxonomy; WII = applied wildlife ecology (पारिस्थितिकी) and training.

> **Logic:** ZSI = **animals only**. Younger than BSI (1890 vs 1916).

28.3 Forest Survey of India (FSI) (भारतीय वन सर्वेक्षण)

Identity: 1981 | Dehradun | ISFR biennial | Forest cover ≥10% canopy

- **FSI** monitors India's **forest and tree cover (वृक्ष आवरण)** using satellite imagery and ground verification.
- Established **1981**; HQ **Dehradun**; under **MoEFCC**.
- Publishes **India State of Forest Report (ISFR)** every **two years** — not annual.
- Forest cover = land **≥1 ha** with canopy density **≥10%**. Classes: VDF (≥70%), MDF (40–70%), OF (10–40%).
- **ISFR 2023:** India forest cover **21.76%**; UP **6.24%**.
- UP districts with **>20%** forest: **Chandauli** and **Shravasti** only — **Bahraich is not**.
- Also runs satellite **forest fire alerts**. Does **not** set pollution standards (CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)) or do taxonomy (BSI/ZSI).

PYQ — UPPCS Prelims 2025, Q8

ISFR 2023 — UP districts with forest over >20% of area: (1) Bahraich (2) Chandauli (3) Shravasti

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

Show answer

Ans: C — Chandauli + Shravasti. Bahraich ~10% despite Terai location.

> **Logic:** ISFR = **FSI only**. Bahraich Terai ≠ automatically >20% cover.

28.4 Forest Research Institute (FRI) (वन अनुसंधान संस्थान)

Identity: 1906 | Dehradun | Forestry research | ICFRE | ≠ FSI

- **FRI** is India's premier **forestry research** institute — silviculture, pathology, entomology, wood science, NTFPs.
- Established **1906** (Imperial Forest Research Institute); now under **ICFRE**; HQ **Dehradun**.
- Trains forestry scientists; iconic (साकार) Dehradun campus.
- **FRI ≠ FSI:** both in Dehradun — FRI = **how to grow/manage forests**; FSI = **how much forest exists on the map**.

- FRI does **not** publish ISFR.

Feature	FRI	FSI
Year	1906	1981
Role	Research	Cover survey
Output	Silviculture protocols	**ISFR**

> **Logic:** "FRI publishes ISFR" = **FALSE**.

28.5 Wildlife Institute of India (WII) (भारतीय वन्यजीव संस्थान)

Identity: 1982 | Dehradun | Wildlife research + IFS training | Tiger census protocols

- **WII** is MoEFCC's autonomous institute for **wildlife research, training, and conservation science**.
- Established **1982** at **Dehradun**.
- Trains **IFS** and state forest officers in wildlife management.
- Designs **All India Tiger Estimation** scientific protocols (with NTCA).
- Studies population (जनसंख्या) ecology, human–wildlife conflict, corridors (including UP Terai).
- **Does not** do national plant/animal taxonomy (BSI/ZSI) or publish ISFR (FSI).

> **Logic:** Dehradun trio — **FSI** (cover), **FRI** (forestry research), **WII** (wildlife).

28.6 National Biodiversity Authority (NBA) (राष्ट्रीय जैव विविधता प्राधिकरण)

Identity: BDA 2002 | Chennai HQ | Foreign access + ABS | ≠ PBR preparer

- **NBA** is the national statutory body under the **Biological Diversity Act, 2002**.
- Headquarters: **Chennai, Tamil Nadu (नाडु) — not New Delhi**.
- Approves **foreign** access to Indian biological resources; regulates biopiracy; ensures **Access and Benefit-Sharing (ABS)**.
- Implements India's CBD / Nagoya Protocol obligations.
- Advises the Centre on biodiversity policy; can support **Biodiversity Heritage Sites**.
- **Does not** prepare People's Biodiversity Registers — that is **BMC**.

Level	Body	Seat	Role
National	NBA	**Chennai**	Foreign access, ABS
State	SBB	State capital	Domestic commercial access
Local	BMC	Panchayat/ward	**PBR**

> **Logic:** NBA HQ = **Chennai**. "NBA prepares PBR" = **FALSE**.

28.7 State Biodiversity Boards (SBB)

Identity: BDA 2002 | One per state | Domestic access | BMC supervision | ≠ SPCB

- Every state/UT has an **SBB** under BDA 2002.
- **Uttar Pradesh State Biodiversity Board (UPSBB)** sits at **Lucknow** (लखनऊ).
- Regulates **Indian** commercial access to bio-resources within the state; foreign cases go to NBA.
- Guides BMCs; receives and validates **PBRs**; maintains state biodiversity database.
- Can identify/recommend (सिफारिश) **Biodiversity Heritage Sites**.
- **SBB ≠ SPCB:** SBB = biodiversity (BDA); SPCB = pollution (Water/Air Acts).

> **Logic:** Do not confuse State Biodiversity Board with State Pollution Control Board.

28.8 Biodiversity Management Committees (BMC)

Identity: Local BDA body | Prepares PBR | Panchayat/ward level

- **BMC** is the **local** committee under BDA 2002 — at panchayat, municipality, or cantonment level.
- Core duty: prepare and update the **People's Biodiversity Register (PBR)** (जन जैव विविधता रजिस्टर) — local species, habitats, and traditional knowledge.
- Supports local conservation and channels ABS benefits when bio-resources are commercially used.
- Works under SBB guidance; national reporting links upward to NBA.
- **PBR is not prepared by NBA or BSI**.

> **Logic:** BMC → PBR**. National apex ≠ local register.

28.9 National Bureau of Plant Genetic Resources (NBPGR)

Identity: 1978 | New Delhi (Pusa) | ICAR | Plant gene bank | Ex-situ (स्थानांतरित)

- **NBPGR** conserves **plant genetic resources** — seeds, tissue, DNA — for food and agriculture.
- Established **1978**; HQ **New Delhi (Pusa Campus)**; under **ICAR**.
- Operates the **National Gene Bank** (long-term base collection) and active collections for breeders.
- **Ex-situ** (स्थानांतरित) conservation — complementary to in-situ (स्थान पर) protected areas.
- **NBPGR ≠ BSI:** BSI surveys/taxonomy (Kolkata); NBPGR stores germplasm (Delhi).
- **NBPGR ≠ NBAGR:** animals/livestock gene bank is **NBAGR, Karnal**.

> **Logic:** NBPGR = **New Delhi**, plants. Not Kolkata. Not animal gene bank.
